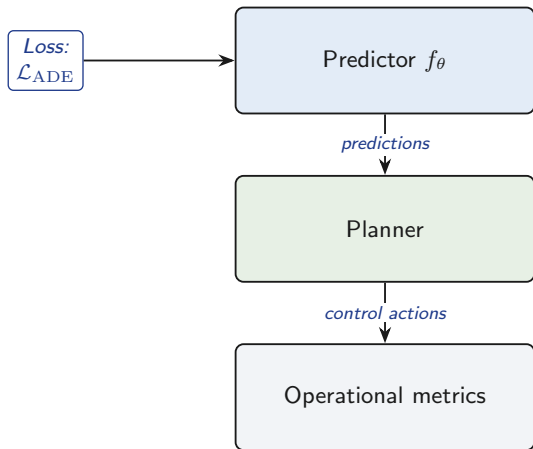


(a) Prediction-Isolated

conventional pipeline

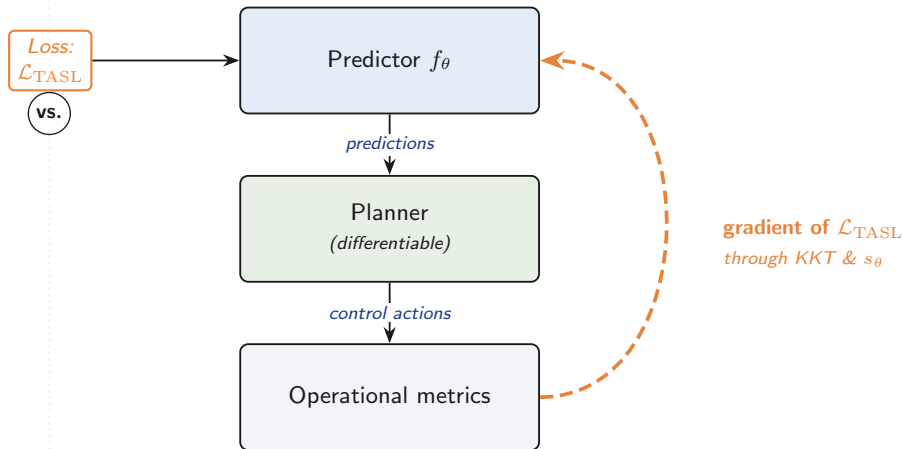


*Predictor trained against displacement;
planner adapts to whatever it receives.*

No gradient flows back.

(b) Joint Formulation

this paper



*Operational signals **supervise the predictor**.
Predictor learns where uncertainty matters most
for downstream safety.*